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Classification of Digits using HOSVD and Tangential Distance

Zagreb, March 2025.

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1 Abstract

In this paper we apply several algorithms for classifying handwritten digits; we work on USPS (United States Postal Service) data. We use several algorithms: first based on HOSVD, second on tangential distances, and the rest are combinations of the previous ones. We compare the obtained accuracies on graphs. Additionally, we use data preprocessing methods; specifically, Gaussian blurring.

2 Introduction

Classification of handwritten digits represents an important problem in the field of computer vision and machine learning, with applications in optical character recognition (OCR), automatic document processing, and authentication systems. In this paper, we explore different approaches to digit classification using the USPS (United States Postal Service) dataset, which contains handwritten numbers of various styles and variations.

We apply multiple classification methods, relying on HOSVD techniques and tangential distances, as well as their combinations. Additionally, to improve model accuracy, we use data preprocessing methods, including Gaussian blurring, which can help reduce noise.

The goal of this paper is to analyze and compare the performance of these methods and present the results obtained through visualizations and comparative analyses of classification accuracy.

3 Definitions and Ideas

3.1 Tensors

Definition 1 (Tensor). *If a field $\mathcal{X}$ is determined by N indices then $\mathcal{X} \in \mathbb{R}^{I_1 \times \dots \times I_N}$ is a tensor of order N over $\mathbb{R}$, $\mathcal{X} = (x_{i_1 i_2 \dots i_N})$, $i_1 = 1, \dots, I_1; \dots; i_N = 1, \dots, I_N$.*

Definition 2 (Fiber of tensor mode k). *Let a tensor $\mathcal{X} \in \mathbb{R}^{I_1 \times \dots \times I_N}$ of order N over $\mathbb{R}$, $\mathcal{X} = (x_{i_1 i_2 \dots i_N})$, $i_1 = 1, \dots, I_1; \dots; i_N = 1, \dots, I_N$ be given. If we fix all indices except k -th, then we call that part of the tensor a fiber of tensor mode k .*

3.2 HOSVD - SVD of Tensors

Let $B \in \mathbb{R}^{m \times n}$, $m \geq n$; $B = U \Sigma V^T$, $U^T U = I_m$, $V^T V = I_n$,

$$\Sigma = \text{diag}(\sigma_1, \dots, \sigma_n); \sigma_1 \geq \dots \geq \sigma_r \geq \sigma_{r+1} = \dots = \sigma_n = 0$$

$$\begin{pmatrix} * \\ * \end{pmatrix} = \begin{pmatrix} + & + & + & + & + & + & + & + & + & + & * & * & * \\ + & + & + & + & + & + & + & + & + & + & * & * & * \\ + & + & + & + & + & + & + & + & + & + & * & * & * \\ + & + & + & + & + & + & + & + & + & + & * & * & * \\ + & + & + & + & + & + & + & + & + & + & * & * & * \\ + & + & + & + & + & + & + & + & + & + & * & * & * \\ + & + & + & + & + & + & + & + & + & + & * & * & * \end{pmatrix} \begin{pmatrix} * \\ * \end{pmatrix} \begin{pmatrix} xxxxxx \\ xxxxxx \\ xxxxxx \\ xxxxxx \end{pmatrix}$$

$$B_k = U(:, 1:k) \Sigma(1:k, 1:k) V(:, 1:k)^T; \|B - B_k\|_2 = \sigma_{k+1}$$

$$\begin{pmatrix} * \\ * \end{pmatrix}_{m \times n} \approx \begin{pmatrix} + & + & + & + \\ + & + & + & + \\ + & + & + & + \\ + & + & + & + \end{pmatrix}_{m \times k} \begin{pmatrix} * \\ * \end{pmatrix}_{k \times k} \begin{pmatrix} xxxxx \\ xxxxx \end{pmatrix}_{k \times n}$$

Analogous reasoning can be applied to tensors. Consider the following theorem.

Theorem 1. *Tensor $A \in \mathbb{R}^{\ell \times m \times n}$ can be written as*

$$A = S \times_1 U^{(1)} \times_2 U^{(2)} \times_3 U^{(3)}$$

where $U^{(1)} \in \mathbb{R}^{\ell \times \ell}$, $U^{(2)} \in \mathbb{R}^{m \times m}$, $U^{(3)} \in \mathbb{R}^{n \times n}$ are orthogonal matrices, S is a tensor with the same dimensions as A and has the property of complete orthogonality: any two different slices of the same type are orthogonal: for all $i \neq j$ it holds

$$\langle S(i, :, :), S(j, :, :) \rangle_F = \langle S(:, i, :), S(:, j, :) \rangle_F = \langle S(:, :, i), S(:, :, j) \rangle_F = 0.$$

Singular values in mode 1 are

$$\sigma_j^{(1)} = \|S(j, :, :)\|_F, j = 1, \dots, \ell$$

and are ordered such that $\sigma_1^{(1)} \geq \sigma_2^{(1)} \geq \dots \geq \sigma_\ell^{(1)} \geq 0$. If $\ell > mn$, then $S(i, :, :) = 0$ for $i > mn$.

Analogously, singular values in other modes are defined.

4 Algorithms

4.1 HOSVD

Let us recall our main theorem again.

Theorem 2 (HOSVD). *Tensor $\mathcal{A} \in \mathbb{R}^{l \times m \times n}$ can be written as*

$$\mathcal{A} = S \times_1 U^{(1)} \times_2 U^{(2)} \times_3 U^{(3)}$$

where $U^{(1)} \in \mathbb{R}^{l \times l}$, $U^{(2)} \in \mathbb{R}^{m \times m}$, $U^{(3)} \in \mathbb{R}^{n \times n}$ are orthogonal matrices, S is a tensor with the same dimensions as $\mathcal{A}$ and has the property of complete orthogonality: any two different slices of the same type are orthogonal: for all $i \neq j$ it holds

$$\langle S(i, :, :), S(j, :, :) \rangle_F = \langle S(:, i, :), S(:, j, :) \rangle_F = \langle S(:, :, i), S(:, :, j) \rangle_F = 0.$$

Singular values in mode 1 are

$$\sigma_j^{(1)} = \|S(j, :, :)\|_F, j = 1, \dots, l$$

and are ordered such that $\sigma_1^{(1)} \geq \sigma_2^{(1)} \geq \dots \geq \sigma_l^{(1)} \geq 0$. If $l > mn$, then $S(i, :, :) = 0$ for $i > mn$. Analogously, singular values in other modes are defined. Also, it holds

$$S = \mathcal{A} \times_1 (U^{(1)})^T \times_2 (U^{(2)})^T \times_3 (U^{(3)})^T.$$

□

It can be easily shown that the decomposition

$$\mathcal{A} = \sum_{i=1}^n \mathbb{A}_i \times_3 u_i^{(3)}$$

holds, where $\mathbb{A}_i = S(:, :, i) \times_1 U^{(1)} \times_2 U^{(2)}$ for which $\langle \mathbb{A}_i, \mathbb{A}_j \rangle_F = 0$ for $i \neq j$. From this representation we can read the frontal slices

$$\mathcal{A}(:, :, j) = \sum_{i=1}^n (u_i^{(3)})_j \cdot \mathbb{A}_i, \quad \forall j = 1, \dots, n$$

Idea: Suppose we have digit samples, for digit k exactly n_k samples. We will create 10 tensors $(\mathcal{A}^{(k)})_{k=0}^9$, one for each digit and we will compute the corresponding *basis* matrices

$\mathbb{A}_i^{(k)}$. Then when it comes time to classify an unknown digit Z , we project that digit onto the spaces $[\{\mathbb{A}_1^{(k)}, \dots, \mathbb{A}_{n_k}^{(k)}\}]$ for $k = 0, \dots, 9$ that is, we compute

$$R(k) = \min_{z_i^{(k)}} \left\| Z - \sum_{i=1}^{n_k} z_i^{(k)} \mathbb{A}_i^{(k)} \right\|_F$$

Since we project onto a space given by an orthogonal basis, we can compute the coefficients explicitly:

$$z_i^{(k)} = \frac{\langle Z, \mathbb{A}_j^{(k)} \rangle_F}{\|\mathbb{A}_j^{(k)}\|_F^2}$$

That is, we cannot do better than these linear approximations so we compute these 10 minima, select the smallest one and declare Z to be that digit. It should be noted that it is desirable to work with orthonormalized bases and a normalized digit $\mathbb{Z}$; then the computation can be performed over the real numbers as:

$$R(k)^2 = 1 - \sum_{i=1}^{n_k} (z_i^{(k)})^2$$

4.2 Tangential Distance

Problem we encountered in digit classification by searching for Euclidean distance between two images is the failure to recognize the correct digit if it is transformed in some way. The most common cases are translation of the digit in some direction or diagonal, small rotations, or thickening. The idea is to create an algorithm that is robust to the mentioned changes. As an example, consider the following image.

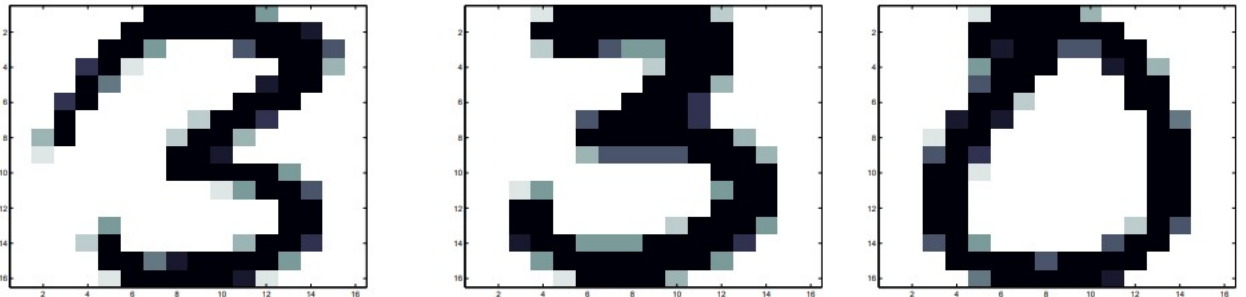


Figure 4.1: Observed digit, Test 1, Test 2

With Euclidean distance we get that Test 2 is closer to the observed digit than Test 1, which is obviously wrong. Motivated by this we introduce tangential distance which makes the initial digit invariant to small changes in rotation, translation, thickening and similar.

Consider the simplest example. 16×16 images can be interpreted as points in $\mathbb{R}^{256}$. Let p represent some pattern on the image. In our case this is a specific digit. Consider the case of only one allowed transformation, translation of the pattern (digit) in the x -direction. This translation can be considered as moving the pattern along a curve in $\mathbb{R}^{256}$. Let the curve be parameterized by parameter $\alpha \in \mathbb{R}$ such that it is given by $s(p, \alpha)$ and $s(p, 0) = p$ holds. Generally, the curve is not linear, but assume it can be approximated by the first two terms of the Taylor expansion for a small value of α :

$$s(p, \alpha) = s(p, 0) + \frac{ds}{d\alpha}(p, 0)\alpha + O(\alpha^2) \approx p + t_p\alpha,$$

where $t_p = \frac{ds}{d\alpha}(p, 0)$ is a vector in $\mathbb{R}^{256}$. Clearly $p + t_p\alpha$ represents the tangent and by changing α slightly around 0, the pattern moves along that tangent at the point p on the curve. Assume we have another pattern e which is approximated in a similar way:

$$s(e, \alpha) \approx e + t_e\alpha.$$

We want to determine the "distance" of these two patterns without taking into account the position of the pattern on the image with respect to the x axis. That is, small moves along curves are allowed, because such small moves should not affect the distance function. Therefore we would ideally like to define a distance measure between p and e as the shortest distance between two curves $s(p, \alpha)$ and $s(e, \alpha)$.

distance between $s(p, \alpha)$ and $s(e, \alpha)$

However, the distance between curves generally cannot be easily calculated, so we can use first-order approximations and calculate the shortest distance between two tangents at points p and e . That is, we will move the patterns independently along their corresponding tangents until we find the smallest distance. If we measure this distance in the standard Euclidean norm, we solve the least squares problem:

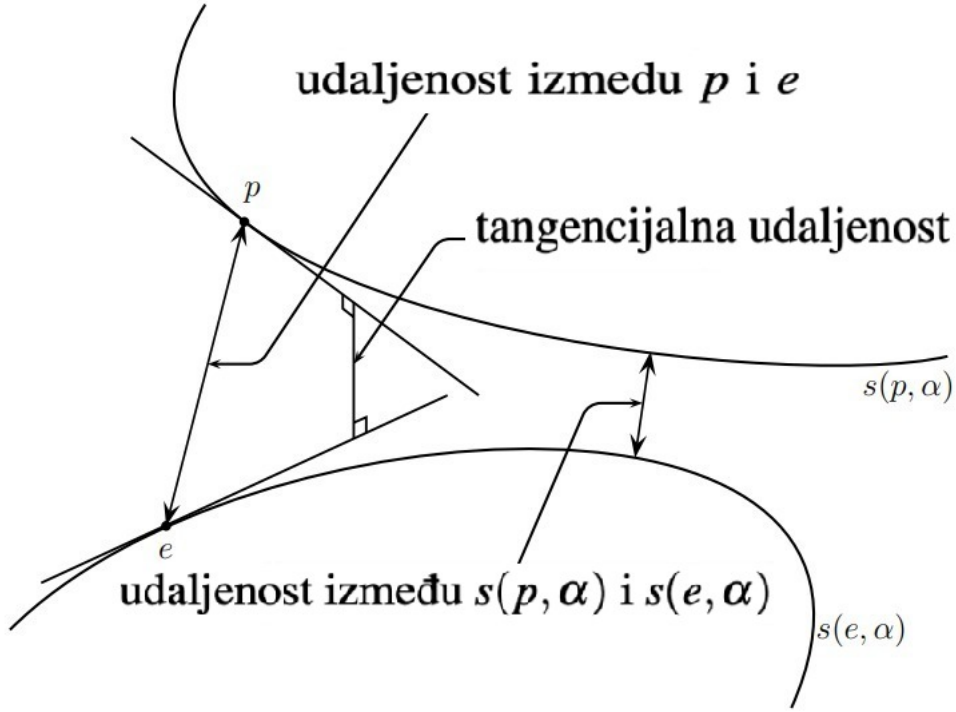


Figure 4.2: Distance of two curves and their tangents

$$\min_{\alpha_p, \alpha_e} \|p + t_p \alpha_p - e - t_e \alpha_e\|_2 = \min_{\alpha_p, \alpha_e} \left\| (p - e) - \begin{pmatrix} -t_p & t_e \end{pmatrix} \begin{pmatrix} \alpha_p \\ \alpha_e \end{pmatrix} \right\|_2.$$

Consider now the case when pattern p is allowed to move along l different curves in $\mathbb{R}^{256}$, parameterized by $\alpha = (\alpha_1 \cdots \alpha_l)^T$. These curves will represent translations, rotations, thickening and the like. Assume we have two patterns, p and e , each of which can move within its allowed transformation surface. Ideally we would like to find the shortest distance between surfaces, but since that is impossible to calculate, we define a new distance measure where we compute the distance between two *tangential planes* of the surface at points p and e .

As before, the tangent plane is given by the first two terms of Taylor expansion: The expansion of function $s(p, \alpha)$ can be written as:

$$s(p, \alpha) = s(p, 0) + \sum_i^l \frac{ds}{d\alpha_i}(p, 0) \alpha_i + O(\|\alpha\|_2^2) \approx p + T_p \alpha,$$

where T_p is the matrix:

$$T_p = \begin{pmatrix} \frac{ds}{d\alpha_1} & \frac{ds}{d\alpha_2} & \cdots & \frac{ds}{d\alpha_l} \end{pmatrix},$$

and the derivatives are computed at point $(p, 0)$.

Therefore, *tangential distance* between points p and e is defined as the smallest possible residual in the least squares problem:

$$\min_{\alpha_p, \alpha_e} \|p + T_p \alpha_p - e - T_e \alpha_e\|_2 = \min_{\alpha_p, \alpha_e} \left\| (p - e) - \begin{pmatrix} -T_p & T_e \end{pmatrix} \begin{pmatrix} \alpha_p \\ \alpha_e \end{pmatrix} \right\|_2.$$

We solved the least squares problem using QR decomposition of matrix $A = \begin{pmatrix} -T_p & T_e \end{pmatrix}$. We note that we are not only interested in the solution but in the norm of the residuals. We can write the problem in the form:

$$\min_{\alpha} \|b - A\alpha\|_2, \quad b = p - e, \quad \alpha = \begin{pmatrix} \alpha_p \\ \alpha_e \end{pmatrix}.$$

$$A = Q \begin{pmatrix} R \\ 0 \end{pmatrix} = (Q_1 \quad Q_2) \begin{pmatrix} R \\ 0 \end{pmatrix} = Q_1 R,$$

Pa je norma reziduala dana s:

$$\min_{\alpha} \|b - A\alpha\|_2^2 = \min_{\alpha} \left\| \begin{pmatrix} Q_1^T b - R\alpha \\ Q_2^T b \end{pmatrix} \right\|_2^2.$$

Since Q is an orthogonal matrix, it holds:

$$\min_{\alpha} [\|(Q_1^T b - R\alpha)\|_2^2 + \|Q_2^T b\|_2^2] = \|Q_2^T b\|_2^2.$$

.

Now we have everything needed if we know the matrix $A = \begin{pmatrix} -T_p & T_e \end{pmatrix}$. Further we will consider the image sample as a function of two variables, $p = p(x, y)$, and show that the derivative of each transformation can be expressed as a differential operator which is a linear combination of derivatives $p_x = \frac{dp}{dx}$ and $p_y = \frac{dp}{dy}$.

4.2.1 Translations

We consider the movement of the pattern using parameter α_x in the x -direction:

$$s(p, \alpha_x)(x, y) = p(x + \alpha_x, y).$$

By differentiating we obtain:

$$\frac{d}{d\alpha_x} (s(p, \alpha_x)(x, y)) |_{\alpha_x=0} = \frac{d}{d\alpha_x} p(x + \alpha_x, y) |_{\alpha_x=0} = p_x(x, y).$$

Analogously, for translation in the y-direction we obtain:

$$\frac{d}{d\alpha_y} (s(p, \alpha_y)(x, y)) |_{\alpha_y=0} = p_y(x, y).$$

4.2.2 Rotation

We rotate the pattern by angle α_r by replacing the value of p at point (x, y) with the value at the point

$$\begin{pmatrix} \cos \alpha_r & \sin \alpha_r \\ -\sin \alpha_r & \cos \alpha_r \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}.$$

Therefore we define the function:

$$s(p, \alpha_r)(x, y) = p(x \cos \alpha_r + y \sin \alpha_r, -x \sin \alpha_r + y \cos \alpha_r),$$

and we obtain the derivative:

$$\frac{d}{d\alpha_r} (s(p, \alpha_r)(x, y)) = (-x \sin \alpha_r + y \cos \alpha_r) p_x + (-x \cos \alpha_r - y \sin \alpha_r) p_y.$$

Evaluation at point $\alpha_r = 0$ gives:

$$\frac{d}{d\alpha_r} (s(p, \alpha_r)(x, y)) |_{\alpha_r=0} = yp_x - xp_y,$$

where the derivatives are evaluated at (x, y) .

4.2.3 Scaling

Scaling the pattern is achieved by defining the following function:

$$s(p, \alpha_s)(x, y) = p((1 + \alpha_s)x, (1 + \alpha_s)y),$$

and we obtain the derivative:

$$\frac{d}{d\alpha_s} (s(p, \alpha_s)(x, y))|_{\alpha_s=0} = xp_x + yp_y.$$

Now that we have two basic algorithms, each with its own strengths and weaknesses, the idea is to somehow combine them to take advantage of the strengths of both while avoiding their weaknesses.

4.3 Hybrid 1.0

In this algorithm we combine the features of HOSVD classification and tangential distance classification; the main idea is to **replace** Euclidean distance with tangential distance; in this way for each digit we try to classify we only need to make 10 comparisons.

1. First we perform the ordinary HOSVD classification algorithm, but instead of computing residuals, we compute orthogonal projections of an arbitrary digit $\mathbb{Z}$; in the notation of the HOSVD section we have $\mathbb{V}^{(k)} = \sum_{i=1}^{n_k} z_i^{(k)} \mathbb{A}_i^{(k)}, k \in \{0, \dots, 9\}$.
2. In the second step, we compute for which $\mathbb{V}^{(k_0)}$ the smallest tangential distance from digit $\mathbb{Z}$ is achieved; we estimate the value of digit $\mathbb{Z}$ as k_0 .

Such an algorithm will be significantly faster than one that is based solely on tangential distances; however, due to the small number of comparisons we expect lower accuracy.

4.4 Hybrid 2.0

The idea of this algorithm is to exploit the speed of the HOSVD algorithm and the accuracy of Tangential Distance. Computing tangential distance is expensive and time-consuming, so we will try to avoid unnecessary comparisons.

1. The first step is to perform an ordinary HOSVD on all digits and look at the confusion matrix to find out which digits the algorithm most often confuses.

As an example, let us read from the table which digits HOSVD assigns to digit 3. For that we look at row 4. The algorithm assigned the digit three to eight 6 times, to five once, and correctly classified it 129 times.

	0	1	2	3	4	5	6	7	8	9
0	313	0	7	1	2	4	2	0	4	0
1	0	207	1	0	1	0	0	1	0	0
2	2	0	160	5	2	0	0	1	2	0
3	0	0	0	129	0	1	0	0	6	0
4	1	0	1	1	168	0	0	2	0	4
5	0	0	7	1	1	108	3	0	0	0
6	0	3	0	0	0	0	143	0	0	0
7	0	0	1	0	2	0	0	115	0	2
8	0	1	0	3	0	2	0	2	133	2
9	1	0	0	2	3	3	0	1	2	137

Table 4.1: Confusion matrix (columns are true digits, rows are classifications)

- Now that we have established what confuses the HOSVD algorithm, we will apply the tangential distance algorithm to the logical choice. That is, in the mentioned example, if HOSVD identified a three, we will compare that digit with tangential vectors only from eights and threes. Thus for each digit obtained we compare with tangential distances to the digits that are most often wrong in the HOSVD algorithm. One more example: If HOSVD recognized digit 5, we will run it once more through the tangential distance algorithm with matrices of digits 2, 5 and 6.

5 Results

We conduct tests on the set of handwritten digits from the US Postal Service. The total amount of data we had was 7291 digits; we give the divisions between test and training in the following sections. For algorithms using HOSVD, we use rank $r = 22$ approximation; in other words the orthogonal spaces for approximating digits are all of dimension 22. Furthermore, we used preprocessing on test data; Gaussian blurring with values $\sigma = 0.9$ with a kernel of 3×3 size.

5.1 HOSVD Algorithm

The following graph of accuracy for the pure HOSVD algorithm tested on the first 1707 digits from the test set is given; on it we see why we choose $r = 22$ approximation:

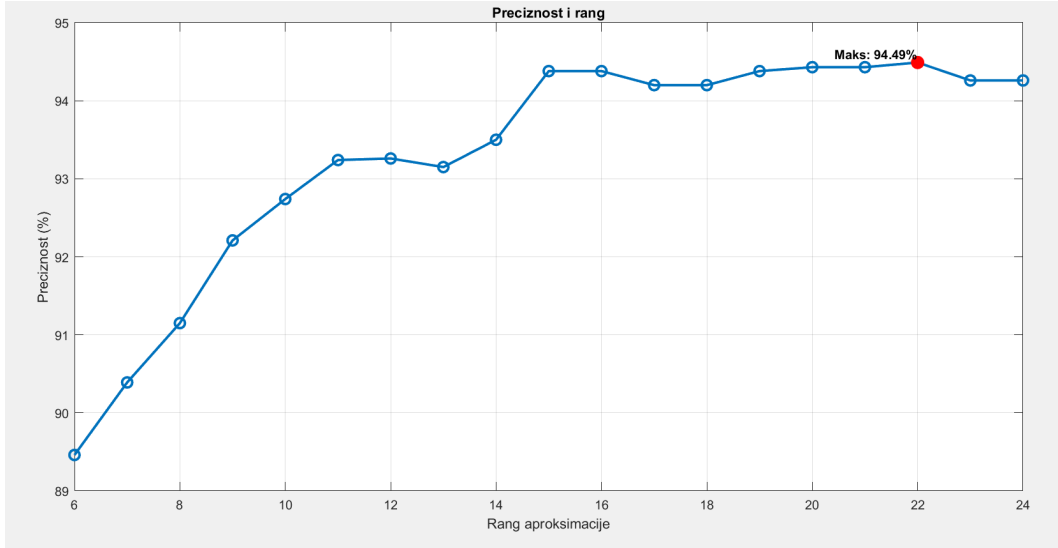


Figure 5.1: Image of the dependence of approximation rank and accuracy

5.2 Tangential Distance

It should be noted that in the tangential distance algorithm we implicitly apply Gaussian blurring; namely, it is necessary to make continuous the discrete structure of the 16×16 matrix P in which we store the digit. Without that we would not be able to determine the tangential vectors. We perform this procedure by convolution with a Gaussian function:

$$g(x, y) = e^{-\frac{x^2 + y^2}{2\sigma^2}}$$

More precisely, the continuous extension of p is given by:

$$P'(x, y) = \sum_{i, j} P[i, j] \delta(x - i) \delta(y - j) \quad (1)$$

$$p(x, y) = P' * g_{\sigma}(x, y) = \sum_{i, j} P[i, j] g_{\sigma}(x - i, y - j) \quad (2)$$

The algorithm is significantly slower and for a smaller percentage is better than HOSVD, which is expected as the concept of tangential distance naturally respects the structure of digits better than pure Euclidean distances.

Table 5.1: Accuracy of tangential distance

(Test, Training)	Success Rate
(100, 600)	0.95
(100, 1000)	0.93
(200, 600)	0.93
(1707, 1707)	0.951

5.3 Hybrid 1.0

Again testing was conducted on 1707 digits; in general we obtained slightly worse results than ordinary HOSVD and tangential distance classification. However, compared to the other algorithm we have significantly greater speed.

5.4 Hybrid 2.0 - Model ChatDPM

Table 5.2: Accuracy of Hybrid 2.0

(Test, Training)	Success Rate
(1707, 1707)	0.92

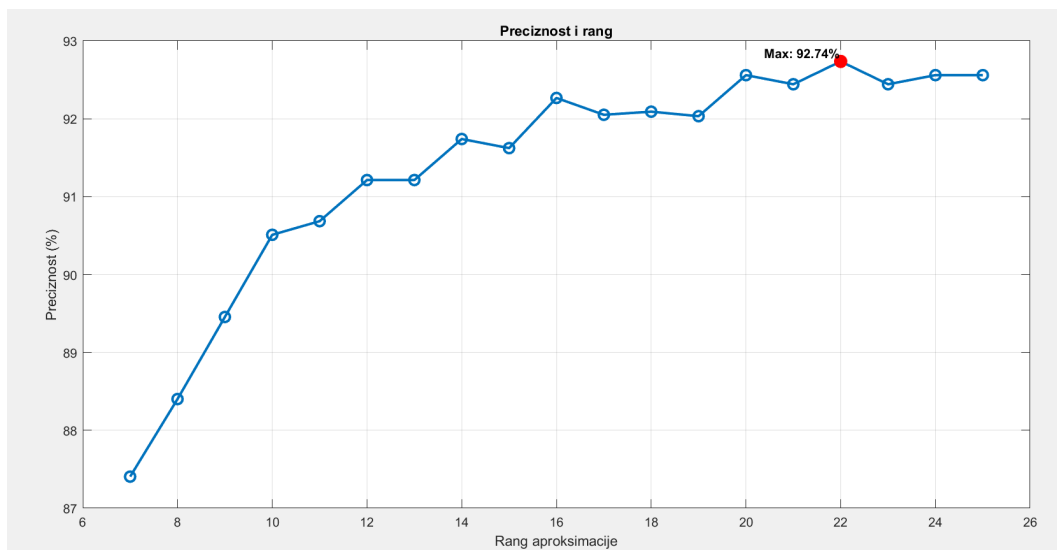


Figure 5.2: Image of the dependence of approximation rank and accuracy

Remark 1 (Extended Dataset). It should be noted that we tested tangential distances on an extended dataset, a total of 7291 digits. We divided this number into 2000 digits that we classified and 5291 digits with which we compared them. Probably due to better handwriting of people, the result is an incredible 98%! The same result is obtained if we work on our standard (100,600) or on the entire dataset.

6 Conclusion

We successfully implemented digit classification algorithms using HOSVD and tangential distances and gave two examples of algorithms that combine both approaches. We tested on limited and extended sets of handwritten digits from the USPS. Particularly for further research it would be worthwhile to come up with methods for optimal choice of rank for each individual digit and with additional ways to speed up the tangential distance method.

7 References

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